

MP352 – Special Relativity

Problem Set 5

Due by the end of lecture on Friday, 8 May 2015

1. Just to get you used to the magnitudes of the units we use in relativistic kinematics, compute the following:
 - (a) The mass, in TeV/c^2 , of a man weighing $11\frac{1}{2}$ stone.
 - (b) The rest energy, in joules, of a mouse of mass 20 g.
 - (c) The momentum, in MeV/c , of a football of mass 0.42 kg moving at $5\text{ m} \cdot \text{s}^{-1}$.
2. This is a classic problem in relativistic kinematics: suppose a electron and a positron annihilate each other in a burst of photons (as usually happens when a particle meets its antimatter evil twin). Show that $e^- + e^+ \rightarrow \gamma$ is forbidden, but $e^- + e^+ \rightarrow \gamma + \gamma$ can occur. (The electron and positron have the same mass $m_e = 0.51\text{ MeV}/c^2$ and the photon is, of course, massless.)
3. The most common decay mode for a positive pion is into an antimuon and a muon-neutrino: $\pi^+ \rightarrow \mu^+ + \nu_\mu$.
 - (a) If the neutrino is massless, show that the magnitudes of the neutrino and antimuon momenta in the centre-of-mass (CM) frame are both

$$|\vec{p}| = \frac{m_\pi^2 - m_\mu^2}{2m_\pi} c.$$

- (b) The neutrino, being massless, travels at the speed of light; how fast is the muon moving? Use $m_\pi = 139.57\text{ MeV}/c^2$ and $m_\mu = 105.66\text{ MeV}/c^2$.
4. A neutron star forms when the temperature in a collapsing star becomes high enough that the electrons can penetrate into the nuclei of the star's atoms, creating neutrons via the reaction $e^- + p \rightarrow n + \nu_e$. To estimate the minimum temperature needed for this to occur, do the following:
 - (a) Assume that each atom in the star is in a frame where the protons are at rest and each electron has total energy E . Find the threshold energy for the reaction in this frame. (Take the neutrino to be massless and use $m_p = 938.27\text{ MeV}/c^2$ and $m_n = 939.57\text{ MeV}/c^2$.)
 - (b) The kinetic energy of each electron is $E - m_e c^2$; if we assume that the star is at a temperature T , then the equipartition theorem says that this kinetic energy must equal $3k_B T/2$. Using $k_B = 8.62 \times 10^{-5}\text{ eV} \cdot \text{K}^{-1}$, determine the minimum T at which a neutron star can begin forming.